

Positive Recurrence of Single-Linkage Bimolecular Weakly Reversible Stochastic Reaction Networks

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Abstract

We prove that every finite bimolecular weakly reversible stochastic mass-action network with one linkage class is positive recurrent on every closed communicating class, for every positive rate vector. No pure-species complex assumption is imposed. The proof augments the embedded jump chain by the target complex of the most recent reaction and removes that target from the population. A residual factorial potential then has the exact increment $\log((x)_t/(x)_s)$ when the next source is s and the carried target is t . Following the carried target has zero reward. A finite library of target-following episodes reduces every possible hierarchy of source propensities to a scalar concave recursion. A normalized-log compactification of an arbitrary divergent sequence then gives either an enabled source of strictly greater asymptotic weight at a lower terminal complex, or an exact reaction-wise conservation law contradicting divergence in the fixed class. This yields one common proper random-time Foster function, a finite exceptional set, and bounded jump-count episodes. A self-contained finite trace-chain argument converts the certificate to finite mean return time for the original continuous-time chain.

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1 Statement and reductions

Let the species set be finite. A complex is a vector $y \in \mathbb{N}_0^d$, and a reaction is an ordered pair $y \rightarrow y'$. The network is *bimolecular* when $|y| \leq 2$ for every complex. Its stochastic mass-action propensity at $x \in \mathbb{N}_0^d$ is

$$\lambda_{yy'}(x) = \kappa_{yy'}(x)_y, \quad (x)_y = \prod_i (x_i)_{y_i},$$

with positive rate constants. We use the inherited fact that every bimolecular weakly reversible system is nonexplosive.

Theorem 1.1 (Single-linkage positive recurrence). *Let a finite bimolecular stochastic mass-action reaction network have one linkage class, and suppose its reaction graph is strongly connected. For every positive rate vector, every closed communicating class of its continuous-time Markov chain is positive recurrent.*

A reaction $y \rightarrow y$ makes no state change and contributes zero to the generator. We discard such reactions throughout. Duplicate reactions may be combined by adding their rates. If the complex graph has one vertex, all closed classes are singletons. If all complexes have the same molecularity, every reaction preserves $\sum_i x_i$, so every communicating class is finite. Thus the only nontrivial case has at least two complexes and an infinite closed irreducible class Γ . Strong connectivity then gives every complex a nontrivial outgoing reaction.

Write

$$\bar{\kappa}_y = \sum_{y \rightarrow u, u \neq y} \kappa_{yu}, \quad \bar{\kappa}_- = \min_y \bar{\kappa}_y, \quad \bar{\kappa}_+ = \max_y \bar{\kappa}_y.$$

All three quantities are finite and the first minimum is positive.

2 The target-augmented jump chain

Consider the embedded jump chain of the CTMC. After a reaction fires, record its target complex. A reachable augmented state is denoted (x, t) , where $x \in \Gamma$, $t \in \mathcal{C}$, and $x \geq t$. Put

$$r = x - t.$$

If the next reaction is $s \rightarrow u$, then

$$(x, t) \mapsto (x - s + u, u), \quad r \mapsto x - s.$$

The transition probabilities depend on x and not on the historical label t , so the augmented process is a Markov chain.

Let $\tilde{\Gamma}$ be the set of augmented pairs reachable after a genuine jump.

Lemma 2.1 (Augmented irreducibility). *The augmented embedded chain on $\tilde{\Gamma}$ is irreducible.*

Proof. Take $(x, t), (x', t') \in \tilde{\Gamma}$. By reachability of the second pair, there is a reaction $s' \rightarrow t'$ and a predecessor population $z = x' - t' + s' \in \Gamma$ from which that reaction is enabled. Population irreducibility gives a finite positive-probability reaction path from x to z . The same path can be followed from the first augmented pair, because augmented labels do not affect propensities. Firing $s' \rightarrow t'$ then reaches (x', t') . $\square$

Define the residual factorial potential

$$V(x, t) = \sum_{i=1}^d \log((x_i - t_i)!). \quad (1)$$

It is nonnegative. Since $\log n! \rightarrow \infty$ and t belongs to a finite set of complexes, every sublevel set of V in $\tilde{\Gamma}$ is finite.

Lemma 2.2 (Exact target/source identity). *At an augmented state (x, t) , if the next reaction has source s and target u , then*

$$V(x - s + u, u) - V(x, t) = \log \frac{(x)_t}{(x)_s}. \quad (2)$$

In particular, a reaction sourced at the carried target, $s = t$, has exactly zero residual reward.

Proof. The new residual is $x - s$. Hence

$$\exp(\Delta V) = \frac{\prod_i (x_i - s_i)!}{\prod_i (x_i - t_i)!} = \frac{(x)_t}{(x)_s}.$$

□

This identity is the reason for augmenting by a target. A product complex is kept as a complete credit until the chain switches to a different source.

3 A one-jump source-probability bound

At a population state x , define

$$\Lambda(x) = \sum_{y \in \mathcal{C}} \bar{\kappa}_y(x)_y, \quad p_x(y) = \frac{\bar{\kappa}_y(x)_y}{\Lambda(x)}$$

for enabled sources. Since the carried target t is present, $p_x(t) > 0$. The expected one-jump residual increment is

$$d(x, t) = \sum_s p_x(s) \log \frac{(x)_t}{(x)_s}.$$

Lemma 3.1 (Entropy bound). *For every reachable augmented state,*

$$d(x, t) \leq \log p_x(t) + C_0, \quad C_0 = \log |\mathcal{C}| + \log \frac{\bar{\kappa}_+}{\bar{\kappa}_-}. \quad (3)$$

Proof. From $(x)_s = p_x(s)\Lambda(x)/\bar{\kappa}_s$ we obtain the exact identity

$$d(x, t) = \log p_x(t) - \sum_s p_x(s) \log p_x(s) + \sum_s p_x(s) \log \bar{\kappa}_s - \log \bar{\kappa}_t.$$

The entropy is at most $\log |\mathcal{C}|$, and the final rate contribution is at most $\log(\bar{\kappa}_+/\bar{\kappa}_-)$. □

No comparison between independent products of rate constants is used. The rates enter only through the fixed additive constant C_0 and positive conditional edge probabilities below.

4 Finite target-following episodes

For every ordered pair $(t, c) \in \mathcal{C}^2$, fix a simple directed path

$$t = y_0 \longrightarrow y_1 \longrightarrow \cdots \longrightarrow y_L = c, \quad L \leq |\mathcal{C}| - 1, \quad (4)$$

which exists by strong connectivity. Choose one particular reaction edge on each step.

Starting from $(r + t, t)$, define the (t, c) -episode as follows. At phase y_k , $k < L$, inspect the next jump. Continue only if the exact designated reaction $y_k \rightarrow y_{k+1}$ fires; otherwise stop immediately after that jump. At terminal phase c , take one further jump and stop.

The designated lift is

$$r + y_0 \longrightarrow r + y_1 \longrightarrow \cdots \longrightarrow r + c.$$

Every designated source is literally present. The residual stays equal to r , and closedness of Γ keeps every lifted state in the class. The episode length is between one and $|\mathcal{C}|$ embedded jumps.

Set

$$q_k = \frac{\kappa_{y_k y_{k+1}}}{\bar{\kappa}_{y_k}} > 0, \quad p_k = p_{r+y_k}(y_k).$$

Let J_k be the expected remaining V -reward from phase k .

Lemma 4.1 (Exact episode recursion). *The expected rewards satisfy*

$$J_L = d(r + c, c), \quad J_k = d(r + y_k, y_k) + q_k p_k J_{k+1}. \quad (5)$$

Proof. The first jump at phase k has expected reward $d(r + y_k, y_k)$. Only the exact designated edge continues. Its probability is $q_k p_k$, and its immediate reward is zero by Lemma 2.2. $\square$

5 Complete-credit elimination

For $\varepsilon \in (0, 1]$, define

$$M_L(\varepsilon) = \log \varepsilon + C_0$$

and, backward along the path,

$$M_k(\varepsilon) = \sup_{0 < p \leq 1} \{ \log p + C_0 + q_k p M_{k+1}(\varepsilon) \}. \quad (6)$$

Lemma 3.1 and induction in Lemma 4.1 give $J_k \leq M_k$ whenever the terminal source probability is ε .

Lemma 5.1 (Scalar elimination). *For every fixed finite path and every fixed positive $q_0, \dots, q_{L-1}$,*

$$M_0(\varepsilon) \longrightarrow -\infty \quad \text{as } \varepsilon \downarrow 0. \quad (7)$$

Proof. For fixed M , the function

$$f(p) = \log p + C_0 + q p M$$

is strictly concave. If $M = -A$ and $qA > 1$, its unique maximizer is $p = (qA)^{-1}$, and

$$\sup_{0 < p \leq 1} f(p) = C_0 - 1 - \log(qA). \quad (8)$$

The terminal bound tends to minus infinity. If $M_{k+1}(\varepsilon) \rightarrow -\infty$, then for all small ε the interior formula (8) applies and also tends to minus infinity. Backward induction proves the claim. $\square$

Corollary 5.2 (Complete-credit episode). *If, along a sequence of residuals, the source probability of the terminal complex c at $r + c$ tends to zero, then the expected V -reward of the entire (t, c) -episode tends to minus infinity.*

This finite recursion absorbs every intermediate source-rate scale. No averaged chain, Schur complement sign assertion, or local Lyapunov switching is needed.

6 Normalized-log compactification

We now prove that every divergent sequence either gives the terminal probability required by Corollary 5.2 or contradicts an exact conservation law.

Let $(x^{(n)}, t^{(n)})$ be a divergent sequence in $\tilde{\Gamma}$. After taking a subsequence, $t^{(n)} = t$ is fixed. Put $r^{(n)} = x^{(n)} - t$. By a finite diagonal extraction, each coordinate of $r^{(n)}$ is either constant or tends to infinity. Let I be the divergent coordinates and $J = I^c$. Define

$$R_n = \sum_{i \in I} \log(r_i^{(n)} + 1).$$

After a further subsequence, the limits

$$w_i = \lim_{n \rightarrow \infty} \frac{\log(r_i^{(n)} + 1)}{R_n}, \quad i \in I, \quad w_i = 0, \quad i \in J, \quad (9)$$

exist. Thus $w \geq 0$ and $\sum_i w_i = 1$. Divergent coordinates with $w_i = 0$ encode all slower subtiers.

For $y \in \mathcal{C}$, put

$$h(y) = w \cdot y, \quad a = \max_{y \in \mathcal{C}} h(y), \quad T = \{y : h(y) = a\}.$$

Lemma 6.1 (Top availability or global conservation). *Along the fixed-class divergent sequence, one of the following holds.*

- (i) *There is a real species-linear conservation law $\ell \cdot X$ whose value tends to infinity along the sequence, contradicting membership in one communicating class.*
- (ii) *There are complexes $s, c \in \mathcal{C}$ with $h(s) > h(c)$ such that, for every sufficiently large n , s is enabled at $r^{(n)} + c$.*

Proof. If every complex is in T , then $w \cdot (y' - y) = 0$ for every reaction. Hence $w \cdot X$ is conserved. Some positive-weight coordinate diverges, giving (i).

Assume $T \neq \mathcal{C}$. If a top complex s contains two particles from I , bimolecularity says it contains no bounded-coordinate particle. It is enabled at $r^{(n)} + c$ for every lower complex $c \notin T$, giving (ii).

It remains to suppose no top complex contains two I -particles. Since the maximum is positive, every top complex contains exactly one I -particle. Let $K \subseteq I$ be the species appearing in top complexes. Each $i \in K$ has $w_i = a$. No complex contains two K -particles, and

$$y \in T \iff q_K(y) := \sum_{i \in K} y_i = 1. \quad (10)$$

Indeed, a complex containing one K -particle already has weight at least a , so it must have exactly weight a ; two would have weight $2a > a$. Conversely every top complex contains a species in K .

If every complex has $q_K = 1$, then $M_K = \sum_{i \in K} X_i$ is conserved and diverges, giving (i). Otherwise there is a lower complex with $q_K = 0$.

If a unary top complex K_i exists, it is enabled over any lower terminal complex, giving (ii). Otherwise each top complex has the form $K_i + D$. The service species D lies in J : if $D \in I$, the top complex would contain two I -particles, contrary to the current case.

If a lower complex c contains a service species D , the corresponding source $K_i + D$ is enabled at $r^{(n)} + c$, and (ii) follows. If no lower complex contains any service species, then

$$M_K - \sum_{D \text{ service}} X_D \quad (11)$$

is zero on every complex: a top complex contains one K -particle and one service particle, while a lower complex contains neither. It is therefore an exact reaction-wise conservation law. Its positive part diverges and its negative-coordinate species all belong to J , hence remain fixed along the subsequence. This gives (i). $\square$

The conservation laws in the proof are global on the entire linkage class; no local phase coboundary is promoted to a species invariant.

Lemma 6.2 (Vanishing terminal probability). *Alternative (ii) of Lemma 6.1 implies*

$$p_{r^{(n)}+c}(c) \longrightarrow 0.$$

Proof. For an enabled fixed bimolecular complex y , coordinatewise comparison with (9) gives

$$\frac{1}{R_n} \log(r^{(n)} + c)_y \longrightarrow h(y). \quad (12)$$

Positive-weight divergent coordinates give their normalized logarithms; zero-weight divergent coordinates contribute $o(R_n)$; fixed coordinates contribute $O(1)$. Both s and c are enabled. Since $h(s) > h(c)$,

$$\frac{(r^{(n)} + c)_s}{(r^{(n)} + c)_c} \longrightarrow \infty.$$

Therefore

$$p_{r^{(n)}+c}(c) \leq \frac{\bar{\kappa}_c(r^{(n)} + c)_c}{\bar{\kappa}_s(r^{(n)} + c)_s} \longrightarrow 0.$$

$\square$

7 Finite-family uniformization

For each $c \in \mathcal{C}$, let $D_c(x, t)$ be the expected V -increment of the fixed (t, c) -episode. Define

$$K = \left\{ (x, t) \in \tilde{\Gamma} : \min_{c \in \mathcal{C}} D_c(x, t) > -1 \right\}. \quad (13)$$

Proposition 7.1 (Finite-flag uniformization). *The set K is finite.*

Proof. If K were infinite, properness of V would give a divergent sequence in K . Apply the compactification above. A conservation alternative in Lemma 6.1 contradicts the fixed communicating class. Otherwise, Lemmas 6.2 and 5.1 give a fixed terminal complex c such that

$$D_c(x^{(n)}, t) \longrightarrow -\infty.$$

This contradicts $(x^{(n)}, t) \in K$, where every terminal episode has drift greater than -1 . $\square$

Outside K , choose before the episode starts the lexicographically first terminal complex minimizing $D_c(x, t)$. This is a deterministic measurable selector on the countable state space. It is not chosen after observing an outcome. The selected stopping time τ satisfies

$$1 \leq \tau \leq L := |\mathcal{C}|, \quad \mathbb{E}_{(x,t)}[V(Z_\tau) - V(x, t)] \leq -1, \quad (x, t) \notin K. \quad (14)$$

There is no defect-promotion gap. The episode has at most L jumps, so every population coordinate changes by at most $2L$. The rank is simply the number of designated path edges remaining: a designated jump decreases it, and every other jump stops the episode.

8 The random-time Foster conclusion

We prove the needed countable-chain statement directly.

Let $Y_0 = z$ and let Y_{n+1} be the endpoint of the episode selected at Y_n . Inside K , define an episode to be one ordinary jump. Put

$$\sigma_K = \inf\{n \geq 0 : Y_n \in K\}.$$

For every integer N , summing the conditional drift inequalities up to the bounded index $N \wedge \sigma_K$ gives

$$\mathbb{E}_z V(Y_{N \wedge \sigma_K}) + \mathbb{E}_z (N \wedge \sigma_K) \leq V(z). \quad (15)$$

Since $V \geq 0$, monotone convergence yields

$$\mathbb{E}_z \sigma_K \leq V(z). \quad (16)$$

The augmented embedded chain reaches K in at most $L\sigma_K$ jumps, so its expected hitting time of K is finite from every state.

We now close the finite-set-to-single-state step explicitly.

Proposition 8.1 (Finite trace-chain closure). *The augmented embedded chain has a state with finite expected return time.*

Proof. For $k \in K$, take one ordinary jump. There are finitely many possible successors, each with finite expected hitting time of K . Hence the next return time to K has finite expectation from every $k \in K$.

Record successive visits to K . The resulting trace chain is finite and irreducible: any augmented path between two points of K , with its intermediate K -visits retained, gives a positive trace path. Fix $k_* \in K$. Finiteness and irreducibility give integers m and $\epsilon > 0$ such that, from every trace state, k_* is reached within m trace transitions with probability at least ϵ . Geometric blocking gives finite expected trace return time to k_* .

Finally set

$$B = \max_{k \in K} \mathbb{E}_k T_K^+ < \infty.$$

Strong Markov conditioning bounds the expected number of original jumps in a random collection of trace excursions by B times the expected number of excursions. The augmented embedded return time to k_* is therefore finite. $\square$

Projecting away the target label shows that the original population embedded chain has finite expected return time to the population coordinate of k_* .

9 Conversion to physical time and proof of the theorem

Let

$$\kappa_* = \min\{\kappa_{yu} : y \neq u\} > 0.$$

At every state of an infinite irreducible class, some nontrivial reaction is enabled. Its falling-factorial factor is a positive integer and hence at least one. Therefore the total CTMC rate is at least κ_* , and every conditional mean holding time is at most $1/\kappa_*$.

If N is the finite-mean embedded return time found above and H_j are the physical holding times, Tonelli's theorem gives

$$\mathbb{E} \sum_{j=0}^{N-1} H_j = \mathbb{E} \sum_{j \geq 0} \mathbf{1}_{\{j < N\}} \frac{1}{\Lambda(X_j)} \leq \frac{\mathbb{E}N}{\kappa_*} < \infty.$$

Nonexplosion excludes accumulation of jumps. Thus one state of Γ has finite expected CTMC return time. Irreducibility gives the same classification for every state, proving Theorem 1.1.

For completeness, every selected episode has at most L jumps and each live phase has total rate at least κ_* . Its physical duration is stochastically dominated by a Gamma random variable with shape L and rate κ_* , so all duration moments are uniformly finite. The coordinate overshoot is deterministically at most $2L$.

10 Scope and verification

The proof automatically respects coordinate faces, siphons, parity classes, and other lattice restrictions because every constructed path consists of actual enabled reactions lifted inside the given closed class. No irreducibility of the whole orthant is assumed.

The one-linkage hypothesis is used in exactly one global place: for every current target and every terminal complex, the finite directed path (4) exists. The argument therefore makes no claim for arbitrary multiple-linkage networks.

The accompanying deterministic verifier checks:

- the exact falling-factorial residual identity on small states;
- the entropy/source-probability rewrite;
- equality of the episode recursion and independent branch enumeration;
- the sharp scalar-envelope calculus;
- the complete top availability/conservation case split;
- an exhaustive atlas of 57,288 three-species complex-set/weight cases;
- 20,000 independent random four-species cases;
- canonical trigger-and-drain, immigration-death, and multi-type calibrations.

None of the finite enumeration is used as a universal proof. The universal steps are the exact identities and Lemma 6.1.

Research-process note

The proof and verification package were developed in an autonomous AI-assisted research workflow. The mathematical claim is presented only through arguments and deterministic checks that can be audited independently.

A Audit against common invalid arguments

The proof does not infer recurrence from nonexplosion, does not normalize an arbitrary stationary measure, does not use deterministic persistence, does not assert one-step negativity of total count or ordinary entropy, and does not invoke complex balance. Reachability alone is not treated as drift: all competing jumps enter the exact finite recursion. The embedded-to-continuous conversion is explicit, and boundary states are handled by the carried-target lift.

The Phase-IV bounded-defect computations are preserved but are not used as a black box. The final target-following certificate explicitly tracks target retention, population feasibility, path containment, positive continuation probabilities, deterministic overshoot, and physical duration.